

Chapter 141 — Contemporary Open Surgery of the Kidney

HISTORICAL PERSPECTIVE

- **Open renal surgery remains an essential tool** despite the shift to minimally invasive techniques: advanced kidney cancers, anatomically complex tumours, and kidney pathology in patients with significant prior abdominal surgery.
- **Technique has evolved with understanding of surgical anatomy.** The **thoracoabdominal and dorsal lumbotomy approaches** are now largely historical because of their morbidity and complexity; they have been supplanted by simpler incisions that give more direct access to the kidney and its vasculature.
- **The kidney lies deep in the retroperitoneum beside the aorta and inferior vena cava (IVC)**, so open surgery demands detailed knowledge of both intraperitoneal and retroperitoneal anatomy.
- **The anterior approach requires displacement of the intraperitoneal organs:** the surgeon must know their vasculature, their attachments to the posterior parietal peritoneum and how they are mobilised, and must keep them out of the field with **self-retaining retractors**.

Concept — open renal surgery has survived as the operation of exposure and vascular control: it is chosen where the kidney must be reached past mobilised viscera with early command of the great vessels.

PREOPERATIVE EVALUATION AND PREPARATION

Laboratory Studies and Renal Function Assessment

- **The goal of almost every renal operation is preservation of global renal function** within the constraints of the disease being treated.
- **Baseline function guides three decisions:** the operation (**partial versus radical nephrectomy**), counselling on the risk of postoperative renal dysfunction, and planning for temporary or permanent dialysis.
- **Urinalysis and serum creatinine (SCr) are obtained in all patients** before renal surgery; the glomerular filtration rate (GFR) is then estimated by the Modification of Diet in Renal Disease (MDRD) equation or the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation — **CKD-EPI is more accurate and now more widely used**.

GFR-ESTIMATING EQUATIONS: MDRD VERSUS CKD-EPI

	MDRD	CKD-EPI (2021, race-free)
Formula	$\text{GFR (mL/min/1.73 m}^2\text{)} = 175 \times \text{SCr}^{-1.154} \times \text{Age}^{-0.203}$	$\text{eGFR} = 142 \times \min(\text{SCr}/\kappa, 1)^\alpha \times \max(\text{SCr}/\kappa, 1)^{-1.200} \times 0.9938^{\text{Age}}$
Sex term	$\times 0.742$ if female	$\times 1.012$ if female; $\kappa = 0.7$ (female), 0.9 (male); $\alpha = -0.241$ (female), -0.302 (male)
Race term	$\times 1.212$ if Black	None
Inputs	SCr (mg/dL), age, sex, race	SCr (mg/dL), age (years), sex
Standing	Older; less accurate	More accurate; preferred

- **The pattern:** both equations rest on creatinine, so both inherit its dependence on muscle mass and diet; CKD-EPI refines the curve with sex-specific inflection points (κ) rather than a single slope.
- **The trap:** κ is the creatinine inflection value (0.7 or 0.9), not an age coefficient; the age term is a single 0.9938^{Age} for both sexes.

FILTRATION MARKERS: CREATININE VERSUS CYSTATIN C

	Creatinine	Cystatin C
Source	Muscle metabolism	Low-molecular-weight protein produced at a constant rate by all nucleated cells
Influenced by	Muscle mass, diet, age, sex	Independent of muscle mass and diet
Renal handling	Filtered	Freely filtered; almost entirely reabsorbed and catabolised by tubular cells — minimal extrarenal elimination
Performance	Insensitive to mild dysfunction	More sensitive to mild change in GFR; superior where creatinine is inaccurate

- **Low risk of postoperative dialysis after open renal surgery:** no comorbidity affecting renal function (hypertension, diabetes) and two normally functioning kidneys.
- **Increased risk: GFR below 60 mL/min or significant proteinuria on urinalysis** → perioperative **nephrology consultation** to optimise function before and after surgery.
- **Poorly controlled diabetes or hypertension risks further deterioration even when GFR is 60 mL/min or above** — record these conditions and ensure they are managed perioperatively.
- **Contemporary open renal surgery is reserved mostly for locally advanced disease (IVC tumour thrombus) or metastatic disease in the cytoreductive setting.** Complete blood count and comprehensive metabolic profile add value for risk stratification — **Memorial Sloan Kettering Cancer Center (MSKCC; Motzer)** and **International Metastatic RCC Database Consortium (IMDC) models** — and for planning.
- **Renal cell carcinoma (RCC) is prone to paraneoplastic phenomena:** haemoglobin and other haematological aberrations aid prognostic stratification; screen advanced disease for **hepatic dysfunction (Stauffer syndrome)** and associated coagulopathy.
- **Haemoglobin assessment matters because major blood loss is a real risk** in advanced disease, especially where transfusion is anticipated.
- **Cardiopulmonary assessment warrants particular care** because intraoperative positioning, blood loss and fluid shifts can cause significant cardiopulmonary compromise.

Concept — renal function assessment is not merely a fitness check: it chooses the operation (partial or radical) and forecasts the need for dialysis, which is why every patient has creatinine, urinalysis and an estimated GFR before any renal surgery.

Preoperative Imaging

- **High-quality cross-sectional imaging of the abdomen — computed tomography (CT) or magnetic resonance imaging (MRI) — is a prerequisite** to any open renal procedure: renal parenchyma, pelvis and ureter, and renal vasculature.
- **It maps the neighbouring organs (colon, pleura, pancreas, spleen, liver)** that may need mobilisation or avoidance or may be involved by disease.
- **Contrast is ideal but may be contraindicated by baseline renal dysfunction.** There is **no universally agreed GFR threshold** (it depends on agent, overall health and context), but **GFR above 30 mL/min is generally acceptable** for both iodinated (CT) and gadolinium-based (MRI) contrast.
- **For locally advanced disease with suspected IVC tumour thrombus, MRI (preferably with intravenous [IV] contrast) or CT venography is preferred:** they best define **cephalad and caudal propagation** and raise suspicion of **caval wall infiltration** — which determine whether liver mobilisation is needed, the level of caval control, and whether a vascular graft or caval resection is required.
- **Thrombus imaging must be current — within 1 month of surgery** — because propagation can be rapid.

- **Three-dimensional (3D) kidney models**, built from CT or MRI data, give an anatomically accurate replica of tumour location, vessels and spatial relationships, allowing the surgeon to explore approaches, rehearse complex steps and anticipate difficulties.
- **Fig. 141.1 (von Hippel–Lindau disease, 4-cm hilar lesion)** shows why each phase is obtained: the noncontrast film shows the lesion, the contrast phase shows enhancement of the mass beside a simple cyst, and the delayed phase shows its proximity to the collecting system.

Renal Artery Embolisation

- **Renal artery embolisation (RAE)** deliberately occludes the arterial supply of the diseased kidney with embolic agents; it is occasionally used before open surgery, mostly for **advanced RCC or large renal tumours**.

PREOPERATIVE ANGIOEMBOLISATION: RATIONALE VERSUS DRAWBACKS

Rationale	Drawbacks
Reduced blood flow → less intraoperative blood loss, haemodynamic support, clearer anatomy	Postinfarction syndrome — flank pain, nausea and fever for up to 72 hours in most patients; often needs systemic steroids
Postembolisation tissue oedema may aid dissection	Retrospective series associate RAE with higher blood loss (oedema of infarcted tissue)
Shrinks large tumours to improve resectability	Harder dissection if surgery is delayed beyond 48 hours
Shrinks arterialised tumour thrombi	Tumour lysis syndrome; embolisation of tumour thrombus ; vascular trauma
Allows renal vein ligation before the artery when large hilar tumours or extensive hilar lymphadenopathy obscure the hilum	Occlusion may be incomplete — inflow must be confirmed intraoperatively
Historically used for large masses, parasitic vessels, tumour thrombi and regional lymphadenopathy	Timing: night before, or ideally the morning of surgery , to limit postinfarction pain

- **The verdict: RAE is used very selectively**; the chapter's authors no longer use prophylactic preoperative angioembolisation even in the most difficult cases, though it remains a tool to know. Its specific place in IVC thrombus surgery is covered under Vena Caval Tumour Thrombectomy.

Prophylactic Measures

- **Surgical site infection (American Urological Association [AUA] guideline): a single dose of a first-generation cephalosporin (cefazolin) or clindamycin, within 1 hour of incision.**
- **Urine culture is not routine**; signs of urinary tract infection prompt culture and sensitivity, and **active infection is treated before surgery**.
- **Mechanical bowel preparation is not indicated** unless intestinal involvement by the renal process is suspected or bowel injury is likely (multiple prior operations, extensive adhesiolysis). If used, weigh **dehydration and acute kidney injury, especially in older adults**.
- **Stress ulcer prophylaxis** with a proton pump inhibitor or sucralfate reduces gastric stress ulcers when prolonged intensive care is expected.
- **Venous thromboembolism (VTE) risk is heightened** by immobilisation and surgical tissue injury, and further by the **hypercoagulable state of malignancy**; intermittent pneumatic compression (IPC) devices reduce postoperative deep venous thrombosis (DVT).

VTE PROPHYLAXIS: GUIDELINE BODIES COMPARED

	American College of Chest Physicians (ACCP)	AUA
Mechanical	—	All patients undergoing open surgery
Pharmacological	Once the bleeding risk has diminished	Consider in patients at elevated DVT risk
Typical regimen for open renal surgery	IPC devices fitted before induction + single preoperative dose of unfractionated heparin 5000 IU subcutaneously	

- **Smokers:** a 4–6-week preoperative cessation programme reduces postoperative complications if time permits.
- **Respiratory prophylaxis:** incentive spirometry for high-risk patients; deep-breathing exercises for low-risk patients.

SURGICAL INSTRUMENTS

- **Appropriate instruments are essential** given the complexity and potential complications of open renal surgery.

Self-Retaining Retractors

- **Purpose:** maintain exposure without handheld retraction, freeing assistants for suction, compression and focused retraction.

SELF-RETAINING RETRACTOR SYSTEMS COMPARED

System	Mounting	Frame	Character
Balfour	Not table-mounted — held by tension against the separated wound edges	—	Usually paired, one at each end of the incision; used more in pelvic than renal surgery; simplest and most rapidly adjustable
Bookwalter	Table-mounted (one or two posts)	Ring over the patient	Stable, more complex; flank blade version used for flank exposure
Omni-Tract	Table-mounted	Bars	Stable, more complex
Thompson	Table-mounted	Bars	Stable, more complex (Fig. 141.3: right retroperitoneal liposarcoma, obese patient, midline incision)

- **The pattern:** stability is bought with complexity — the three table-mounted systems hover blades on bars or a ring; the Balfour trades stability for speed.
- **All systems require experience from surgeon and scrub technician;** persevere through several cases before abandoning a system.

Surgical Clamps

- **For IVC tumour thrombus cases:** two suction units with interchangeable large and small tips; long forceps, right angles and needle drivers (particularly in obese patients).
- **A vascular pan is opened for open renal cases,** with clips, ties and **preloaded Prolene sutures** for prompt vascular suturing.

VASCULAR CLAMPS USED IN OPEN RENAL SURGERY (FIG. 141.4)

Clamp	Configuration / use
Bulldogs	Small occlusion clamps for individual vessels
Satinsky	Curved, side-biting; partial IVC occlusion; en bloc pedicle clamping

Clamp	Configuration / use
Hypogastric	Angled vascular clamp
Glover (duct)	Straight vascular clamp
Coarctation (aortic)	30-degree bend
Renal pedicle clamps (Crawford, Young, Mayo)	Whole-pedicle control

- **Vascular clamps have finely graduated ratchets and an atraumatic jaw surface**, far less injurious to arteries and veins than **nonvascular clamps (Kocher, Shallicross)**.

Energy Devices

- **Kidney, parasitic vessels, liver, adrenal and spleen have different haemostatic needs**, so the surgeon must know each device.

ENERGY DEVICES IN OPEN RENAL SURGERY

Device	Role	Notes
Monopolar cautery	Dissection and haemostasis of vessels under 2 mm	Spray setting for superficial lacerations of liver, adrenal, spleen and kidney ; an insulating sleeve prevents inadvertent arcing along the long skin-to-field trajectory
Bipolar sealers — LigaSure, EnSeal	Vessel sealing	No robust comparative data; perform similarly, EnSeal possibly less radiated heat (anecdotal)
Ultrasonic (Harmonic)	Sealing	Rarely used in open surgery
Irrigated monopolar/ bipolar	Haemostasis with less charring	Adherent char avulsed from the tip can cause more bleeding than was present
Radiofrequency ($\pm$ saline irrigation)	Surface haemostasis	—
Argon beam coagulator	Liver lacerations when other methods fail	—

SURGICAL APPROACHES

- **Adequate exposure is the hallmark of effective open renal surgery**. Proper incision and exposure **minimise the retraction required** and thus retractor-related injury.
- **Landmark structures: right kidney — liver, colon, duodenum; left kidney — spleen, tail of pancreas, colon.**
- **The approach is tailored** to the operation, the anatomy on imaging, previous surgery, body habitus, and limiting factors such as **kyphoscoliosis or pulmonary disease**.

INCISIONS FOR OPEN RENAL SURGERY COMPARED (FIG. 141.6)

Incision	Landmarks	Best for	Limitations
Anterior midline	Xiphoid → around umbilicus → 2 cm below it	Trauma; locally advanced tumours and IVC thrombus ; renovascular; reconstruction (ileal ureter); bilateral procedures	Transperitoneal; bowel mobilisation needed
Anterior subcostal	2 fingerbreadths below costal margin, from just medial to 11th rib tip to medial rectus border	Unilateral transperitoneal access	Divides the abdominal wall muscles

Incision	Landmarks	Best for	Limitations
Chevron (bilateral subcostal)	11th rib tip → below costal margin → curves up in midline → contralateral 11th rib tip	Renovascular surgery; radical nephrectomy with IVC thrombectomy ; both pedicles and great vessels	Extensive muscle division
Subcostal flank	1 fingerbreadth below and parallel to 12th rib, from costovertebral angle	Lower pole, pelviureteric junction, proximal ureter	Poor hilar access — limited for large masses; not ideal for partial nephrectomy
Supracostal flank	Superior border of 11th or 12th rib	Maximal posterior exposure; radical or partial nephrectomy (8-cm mini-flank supra-11th)	Pleural proximity
Transcostal flank	Directly on the 11th rib, toward umbilicus	Retroperitoneal access at the former rib tip	Rib division

Anterior Approaches

Positioning

- **Supine; arms tucked at the side or abducted to 90 degrees** on padded arm boards; **pad the elbows (ulnar nerve)** and all pressure points — **occiput, dorsal torso, sacrum, dorsal legs, heels** — against pressure ulcers and nerve palsy (Fig. 141.5A).
- **Hyperlordosis improves exposure: a rolled blanket under the lumbar spine** (subcostal or chevron) or **table flexion with the umbilicus at the break** (midline).
- **Excess hyperlordosis tensions the great vessels (reducing flow) and causes postoperative low back pain; avoid it in spinal stenosis.**
- **Supine hypotension (aortocaval syndrome)** occurs if adiposity or abdominal masses compress the great vessels; **low back pain** is common with scoliosis or kyphosis; prosthetic hips and knees may be stressed.

ANTERIOR VERSUS FLANK POSITIONING: THE GOVERNING CONTRAST

	Anterior (supine)	Flank (lateral decubitus)
Exposure aid	Lumbar roll or table flexion (hyperlordosis)	Table flexion between iliac crest and costal margin; kidney bar under 12th rib tip
Cardiovascular risk	Aortocaval compression	Reduced pulmonary function and venous return — poor choice in cardiopulmonary disease
Musculoskeletal risk	Low back pain; avoid hyperlordosis in spinal stenosis	Pressure injury, rhabdomyolysis of dependent flank and hip; severe kyphosis limits exposure
Nerve protection	Elbows (ulnar nerve)	Axillary roll (axillary neurovascular bundle)

Anterior Midline Approach

- **Efficient access to both kidneys and the great vessels without muscle division** → less postoperative pain and morbidity.
- **Preferred in renal trauma:** rapid control of the major vessels for haemodynamic stability, and survey of intraperitoneal contents for associated injuries.
- **The authors' preferred approach for locally advanced tumours, including IVC thrombus:** early aortic access permits **early ligation of the renal artery**, reducing blood loss.
- **Technique:** supine, bed slightly flexed with the umbilicus at the break → skin incision **from xiphoid, around the umbilicus, to 2 cm below it** (rarely needed to the pubis) → subcutaneous tissue divided with cautery → **linea alba incised sharply at the midpoint between xiphoid and umbilicus, where preperitoneal fat is thinnest** → finger admitted into the preperitoneal space, incision extended cephalad

and caudally with cautery → preperitoneal fat swept off the peritoneum with atraumatic wide-mouth forceps → **peritoneum grasped with two smooth forceps and incised sharply** → **ligamentum teres divided and suture-ligated** → peritoneal incision lengthened under direct vision.

- **Right kidney exposure: incise the white line of Toldt** (where peritoneal mesothelium reflects over the antimesenteric border of the ascending colon) longitudinally and around the hepatic flexure → with the colon drawn medially, **Toldt's fascia** is dissected bluntly or with delicate cautery to lift the ascending colon and mesocolon off Gerota's fascia → the second part of the duodenum is met and **Kocherised** — **athermally, given the devastating consequences of duodenal injury** → exposure of the right renal vein and lateral IVC → retractor placed; the **right renal artery lies posterior and inferior to the renal vein**.
- **Fig. 141.8:** the ascending and descending mesocolon rest on Gerota's fascia, with **Toldt's fascia interposed**, entered by incising the white line lateral to the colon.
- **Left kidney exposure:** medial reflection of the **descending colon and mesocolon, with the tail of the pancreas superiorly; divide the splenorenal ligament** superolaterally to reflect the spleen medially and expose the upper pole.
- **Cattell-Braasch manoeuvre (right medial visceral rotation)** — direct access to the central vascular compartment (aorta and IVC), especially for right-sided surgery (Fig. 141.7): incise peritoneum lateral to the ascending colon (white line) → continue around the caecum → **Kocherise the duodenum** → incise the posterior parietal peritoneum from the right lower to the left upper quadrant along the root of the small bowel mesentery toward the **ligament of Treitz** → medial and cephalad displacement of the **ascending colon and mesocolon, duodenum (second to fourth parts), jejunum and ileum** with their mesentery. **Landmarks: the left renal vein** (seen as the transverse duodenum is lifted) **or the inferior mesenteric vein (IMV)**.
- **Direct infracolic approach to the renal pedicle** (trauma, or large tumours obscuring the hilum): shift jejunum and ileum aside → **vertical incision in the posterior peritoneum below the ligament of Treitz** → anterior aorta and crossing left renal vein (LRV). **The superior mesenteric artery (SMA) arises from the anterior aorta about 1–2 cm cephalad to the LRV.**
- **The right renal artery can be found in the interaortocaval space at its aortic origin, posterior and inferior to the LRV**, before its retrocaval course → **cephalad retraction of the LRV** allows control and ligation — valuable for **large or medial right tumours** that obscure the artery lateral to the IVC.

Anterior Subcostal Approach

- **Supine; rolled blanket under the lumbar spine** for hyperlordosis.
- **Incision about two fingerbreadths below the costal margin**, from just medial to the tip of the 11th rib to the medial border of the rectus abdominis.
- **Technique:** subcutaneous tissue divided with cautery → **anterior rectus sheath** incised, extended laterally through **external and internal oblique** → rectus divided with cautery → **posterior rectus sheath** divided near the midline → laterally, **internal oblique and transversus abdominis** divided (a right angle may replace the finger as guide) → **transversalis fascia and parietal peritoneum** → peritoneum incised → intraperitoneal mobilisation as for the midline approach → **layered closure of muscle with absorbable sutures**.

Chevron Incision (Bilateral Anterior Subcostal Approach)

- **Ideal for renovascular surgery and radical nephrectomy with IVC tumour thrombectomy**, with good exposure of both renal pedicles and the great vessels.

- **Course:** tip of 11th rib → two fingerbreadths below and parallel to the costal margin → curves superiorly in the midline → parallel to the contralateral costal margin → contralateral 11th rib tip; muscle and sheath division as for the subcostal incision.

Concept — *every anterior exposure is a mobilisation along embryological fusion planes behind the mesocolon: colon and duodenum on the right, colon, pancreas and spleen on the left, and the whole midgut pivoting on the SMA for central access.*

Flank Approaches

Positioning

- **Lateral decubitus with the table flexed between iliac crest and costal margin; kidney bar raised** for retroperitoneal exposure — but beware a **previously repaired contralateral kidney**.
- **Pad all pressure points** to prevent **rhabdomyolysis** of the contralateral flank and hip (Fig. 141.5B).
- **Relative unsuitability: pre-existing cardiopulmonary deficit** (exaggerated decubitus impairs pulmonary function and venous return) and **severe kyphosis** (poor exposure, unexpected pressure on flank and vertebrae) — the surgeon must command alternative approaches.
- **Full set-up (as for the subcostal flank incision):** after induction, endotracheal intubation and **Foley catheter** for urine output → lateral decubitus → **head supported against cervical flexion** → **tip of 12th rib over the kidney bar** → back supported with rolled blanket or beanbag → **dependent leg flexed at hip and knee, top leg straight, pillows between the knees** → **axillary roll just caudal to the axilla** → upper foot padded → nondependent arm on a padded Mayo stand or pillow, horizontal with slight forward shoulder rotation → **flex until the flank muscles are stretched; Trendelenburg until the flank is parallel to the floor** → secure with **2-inch adhesive tape** to the mobile part of the table.

Subcostal Flank Incision

- **Excellent exposure of the proximal ureter and parenchyma** — suited to the **lower pole, pelviureteric junction and proximal ureter**.
- **Hilar access is suboptimal**, limiting it for large masses; **not ideal for partial nephrectomy**, which needs excellent hilar access (Fig. 141.10).
- **Technique:** skin incision begins at the **costovertebral angle at the lateral border of sacrospinalis, just below the 12th rib** → **one fingerbreadth below and parallel to the 12th rib** onto the anterior abdominal wall; **curve gently downward at the midaxillary line to avoid the subcostal nerve**; extend caudally or medially to the lateral rectus border if needed → subcutaneous tissue → cautery through **latissimus dorsi** posteriorly and **serratus posterior inferior** (inserts into the lower four ribs) → **external oblique** anteriorly → exposed **fused lumbodorsal fascia** (origin of internal oblique and transversus) → lumbodorsal fascia and **internal oblique** divided (Fig. 141.13) → two fingers through an opening at the **12th rib tip** sweep the peritoneum medially while **transversus is split digitally** → **subcostal nerve identified between internal oblique and transversus, and spared** (Figs. 141.14–141.15; the subcostal nerve and vessels pierce the lumbodorsal fascia posteriorly and run forward on transversus).
- **More posterior exposure:** incise the posterior angle of the lumbodorsal fascia (sacrospinalis and quadratus lumborum); **dividing the costovertebral ligament lets the 12th rib be retracted superiorly**. A **Bookwalter flank retractor** holds exposure.

Supracostal Flank Incision

- **Above the 11th or 12th rib; favoured by many open renal surgeons.** Turner Warwick (1965) popularised it: **maximal posterior exposure, simpler wound closure, and less morbidity than a transcostal incision requiring rib resection.**

- **An 8-cm modified mini-flank supra-11th rib incision** is a safe, effective approach to radical or partial nephrectomy for renal cortical tumours.
- **Level chosen by anatomy, lesion location and procedure;** positioning as for the subcostal flank.
- **Technique:** skin along the **superior aspect of the 11th or 12th rib**, from the lateral border of sacrospinalis to the lateral border of the rectus → subcutaneous tissue → **latissimus dorsi and serratus posterior inferior** transected posteriorly, revealing the **intercostal muscles** → external and internal oblique divided → **lumbodorsal fascia opened at the rib tip to avoid both peritoneum and pleura** → moving medially, transversus divided while the peritoneum is swept medially and inferiorly → diaphragm exposed by dividing the transversalis layer → **pleura identified between divided transversus and diaphragm, mobilised superiorly** → lateral sacrospinalis incised or retracted to reach the rib neck → **intercostal muscles divided from the distal rib toward the spine, close to the superior border of the rib** to spare the neurovascular bundle; the intercostal nerve is identified and spared → **extrapleural plane developed by entering the investing fascia around the intercostal nerve** → diaphragmatic slips on the inferior ribs transected.

Transcostal Flank Incision

- **Mirrors the supracostal approach, shifted slightly caudally along the 11th rib toward the umbilicus;** the palpable 11th rib is the key landmark.
- **Technique:** lateral incision directly on the 11th rib → subcutaneous tissue and muscle until the anterior surface of the distal 11th rib is seen → **periosteum cleared with a periosteal elevator, protecting the neurovascular bundle beneath the rib** → **distal-most rib divided with a bone cutter;** fragments removed, edges filed with a rongeur, marrow cauterised → remainder as supracostal → **retroperitoneum entered at the former 11th rib tip, avoiding both peritoneal and pleural entry.**

Concept — *the flank incisions are all extraperitoneal and are named by their relation to a rib: the higher the incision, the better the upper-pole and hilar exposure — and the closer the pleura.*

SIMPLE NEPHRECTOMY

- **Removal of the kidney within Gerota's fascia, for nonmalignant disease.**

Indications

- **Permanent poor function** from obstruction, infection, trauma, stones, nephrosclerosis, vesicoureteral reflux, polycystic kidney disease or congenital dysplasia.
- **A well-functioning kidney is rarely removed,** only for intractable pain, bleeding, renovascular hypertension or persistent infection.

Surgical Approach

- **Approach as for radical nephrectomy, but benign disease often distorts anatomy and planes** (prior instrumentation, chronic infection, aberrant anatomy).
- **Perinephric fat is often adherent and inflamed; avoid entering the renal capsule,** which causes bleeding and loss of the plane.
- **Adherent to psoas posteriorly** → include the psoas fascia in the dissection.
- **Large hydronephrotic kidney** → **puncture and aspirate the pelvis** to decompress and aid mobilisation.
- **Spare the adrenal almost universally** by keeping the plane directly on the renal capsule at the upper pole.
- **Partial nephrectomy for benign disease** is occasionally indicated — **hydronephrosis with focal parenchymal atrophy, atrophic pyelonephritis in a duplicated kidney, infected calyceal diverticulum, symptomatic benign tumours such as angiomyolipoma** — using the same technique as for malignancy.

RADICAL NEPHRECTOMY

- **Classic definition (Robson, 1969):** complete removal of the kidney **with Gerota's fascia, the ipsilateral adrenal and a complete regional lymphadenectomy from the crus of the diaphragm to the aortic bifurcation.**
- **Modern practice spares the adrenal when technically feasible** — removal of a radiographically uninvolved adrenal has been associated with worse survival — and reserves **extended lymphadenectomy for selected cases.**

SIMPLE VERSUS RADICAL NEPHRECTOMY

	Simple	Radical
Disease	Benign	Malignant
Plane	Within Gerota's fascia, on the capsule	Outside Gerota's fascia, resected entire
Adrenal	Spared almost universally	Spared if radiographically normal and feasible
Lymph nodes	Not removed	Selective regional lymphadenectomy
Typical difficulty	Inflamed, adherent planes	Size, parasitic vessels, thrombus, adjacent organs

Indications and Workup

- **Radical nephrectomy is reserved for tumours not amenable to partial nephrectomy:** tumour in a **nonfunctioning kidney, large tumours replacing most of the parenchyma, detectable regional lymphadenopathy, consolidative cytoreductive surgery** for metastatic disease, and **renal vein thrombus.**
- **Staging of every suspicious renal tumour: abdominopelvic CT or MRI plus chest imaging** (Fig. 141.18). **The authors prefer chest CT to chest radiography** — plain films miss subcentimetre nodules and mediastinal nodes, RCC favours lung and mediastinum, and missed metastases matter — despite the modest extra radiation.
- **Scrutinise imaging for tumour thrombus, enlarged retroperitoneal nodes, and embryological anomalies** of the collecting system and vasculature.
- **Occult brain metastases occur in a small minority with metastatic RCC; routine brain imaging is not recommended without symptoms.**
- **Routine bone scan is not recommended;** use it for suspicious cross-sectional findings, symptoms, pathological fracture or **raised bone isoenzyme of alkaline phosphatase.**
- **Percutaneous renal biopsy before surgery is considered when:**
 - the patient is uncomfortable with uncertainty about histology;
 - the **false-negative risk is acceptable and a benign result would change the strategy** — a systematic review shows that a substantial minority of patients with a benign biopsy who proceed to surgery have RCC;
 - the mass appears **infiltrative and suspicious for urothelial, metastatic or lymphoid origin;**
 - there is **metastatic disease to be treated with systemic therapy**, where histological subtype directs the agent;
 - there are **bilateral renal tumours** — **synchronous bilateral tumours are more often histologically discordant than metachronous ones;**
 - **clinical trial eligibility** requires it.
- **Preoperative angioembolisation** — historically used for large masses, parasitic vessels, tumour thrombi and hilar lymphadenopathy — is no longer used prophylactically by the authors (rationale and drawbacks tabulated under Renal Artery Embolisation).

- **Large tumours displace organs; distinguishing abutment, displacement and direct invasion (T4) is difficult.** MRI is suggested when CT suggests invasion.
- **True invasion is likelier when the organ changes in appearance, not merely in contour:** tumour infiltration, altered enhancement relative to the rest of the organ, bowel wall thickening. **Suspected involvement of colon, duodenum, liver or spleen → preoperative multidisciplinary planning.**
- **The liver is rarely truly invaded because of its bilaminar capsule,** despite imaging that suggests it. When invasion does occur, depth is usually **under 1 cm**, and **a thin crescent of liver resected with the tumour** suffices — still performed by a liver surgeon.

Surgical Procedure

- **Incisions:** vertical midline, subcostal flank or chevron; retractor Balfour, Bookwalter, Thompson or Omni-Tract; **ribs retracted laterally and cephalad** for upper-quadrant tumours or venous thrombus.
- **Begin with a complete abdominal survey for metastases:** entire bowel, spleen, liver surface, and palpation of the gastric tube in the decompressed stomach.
- *Right radical nephrectomy.* **Liver and gallbladder retracted superiorly;** if more liver mobilisation is needed, the **avascular right triangular ligament** is incised, usually later in the case. **The falciform ligament is left intact when liver mobilisation is not required** — a falciform flap can buttress a duodenal repair — but an intact falciform demands care during retraction because it predisposes to **traction lacerations of the liver.**
- **Sequence:** posterior parietal peritoneum incised **2 cm medial to the white line of Toldt**, from the pelvis (iliac artery) to the right upper quadrant (hepatic flexure) → **anterior pararenal space** developed between the anterior renal fascia and the ascending mesocolon → hepatic flexure mobilised sharply and bluntly → **second part of the duodenum Kocherised medially** (Fig. 141.19) — with extreme care for medial tumours → **IVC identified posteriorly;** dissection anterior to it reveals the **renal vein and right gonadal vein** → vessel loop on the renal vein for gentle traction; **renal vein palpated for thrombus** → **renal artery identified posterior to the vein.**
- **Finding the plane:** mesenteric fat is **vibrant yellow or orange;** Gerota's fascia is **paler** and less saturated; **anterior tension on the colonic mesentery** reveals the areolar plane. Large inflammatory masses obscure it through **neovascularity, lost planes and thickened tissue.**
- **Protect the ascending mesocolon** — injury to the **right colic and ileocolic arteries** devitalises colon. **Bipolar sealing of small neovascular perforators** saves time later.
- **Resect the renal fascia in its entirety** (best chance of cure, no intra-abdominal tumour spillage), keeping the **lymph node plane posterior on the downside.**
- **If the artery is hard to find:** go to the lower pole, identify **ureter and gonadal vein;** spare the gonadal vein if feasible — large tumours often make this unsafe because of **avulsion from the IVC (right) or LRV (left)** → ligate the ureter and **lift the kidney from posterior to anterior** to display the artery behind it.
- **Interaortocaval option for difficult hilar dissection** (Fig. 141.20: the anteromedial IVC leads to the short right renal vein; the artery lies deep to the vein): control the **right renal artery at its aortic take-off** → **ligate with 0 silk (surgical clip in emergencies)** → kidney and tumour **shrink and decompress** → now-flaccid **right renal vein examined for thrombus, doubly ligated (0 silk tie + 3-0 Prolene suture ligature) and divided** → artery lateral to the IVC, now easy to reach, doubly ligated and divided.
- **Lumbar veins.** **Avulsed lumbar veins → 2-0 or 3-0 silk ties. No clips on the IVC side** — they obstruct urgent IVC clamping; well-placed clips on the posterior (lumbar) side before release are acceptable. **Uncontrolled lumbar veins retract** lateral to the vertebra and medial to psoas, causing inaccessible retroperitoneal bleeding → **compress psoas with a sponge stick** at the site to slow bleeding and reidentify

the vein → grasp with **DeBakey forceps** and **apply high-power fulguration** (small lumbar) or place sutures → haemostatic packing plus psoas compression as adjuncts.

- **Left radical nephrectomy. Sequence:** white line of Toldt incised from the **splenic flexure to the common iliac artery** → descending colon reflected medially → **renocolic ligament divided, protecting the tail of the pancreas** → plane developed between the anterior leaf of Gerota's fascia and the **pancreatic and splenic fascia, which is left intact**; a folded moist laparoscopic sponge pads the pancreas from retractor blades → **LRV identified using the anterior surface of the aorta as guide**; **left renal artery usually cranial and posterior to the LRV** → at the lower pole, **left ureter and gonadal vein** identified; the **gonadal vein traced to its LRV insertion** → gonadal vein preserved or ligated according to tumour size and site (ligation rarely causes **gonadal pain, congestion or thrombosis**) → ureter divided, inferior and posterior kidney mobilised to display the artery.
- **Left renal artery division:** fine scalpel; **proximal end — heavy silk (0 or larger) plus 3-0 Prolene suture**; **distal end — 0 silk. Confirm complete visual occlusion of the stump** in the elderly or those with peripheral arterial disease (calcification) → renal vein then secured and divided similarly.

RIGHT VERSUS LEFT RADICAL NEPHRECTOMY

	Right	Left
Visceral mobilisation	Liver/gallbladder up; ascending colon; Kocher manoeuvre	Descending colon; renocolic ligament; pancreatic tail and spleen protected
Guide to the vein	Anterior IVC → short right renal vein	Anterior aorta → long LRV
Artery position	Posterior to vein; interaortocaval at origin	Cranial and posterior to LRV
Gonadal vein	Drains into IVC (avulsion risk from IVC)	Drains into LRV — a guide to the LRV
Organ at risk	Duodenum, liver	Pancreatic tail, spleen; SMA mistaken for the renal artery

- **Whole-pedicle clamping.** When artery and vein **cannot be separated because of hilar lymphadenopathy**, the pedicle is controlled en bloc. **Arteriovenous fistula (AVF) after en bloc ligation is a theoretical risk but extremely rare**; small series of en bloc stapling found none.
- **Sequence:** pedicle bluntly dissected to a **2–3-cm diameter** → **long curved vascular clamps (Satinsky) or pedicle clamps (Crawford, Young, Mayo)** → **first clamp at the proximal-most pedicle**, extending posteriorly beyond all structures to engage the suture → **second clamp just distal and adjacent** under vision → **third clamp near the parenchyma** → **divide between second and third clamps**, removing the specimen with vascular stumps protruding → **0 silk looped below the lower clamp; tie the pedicle twice and add a suture ligature** (silk ties may slip) → where artery and vein stumps are not individually visible, a **running horizontal mattress 3-0 Prolene** beneath the proximal clamp, distal to the silk tie.
- **Figs. 141.23–141.24** contrast the two hilar methods: "**cut first, ligate second**" versus "**ligate first, cut second**".
- **Loss of control of the hilar pedicle.** **Stay calm** → **inform the anaesthetist and all theatre staff** → **aggressive hydration and blood products** → **compression with a fingertip or sponge stick** (also on the IVC and/or aorta) while the team prepares → **two Yankauer suckers** clear the field → **never clamp blindly: suction, pack, retract and dissect for exposure**, then clamp and ligate with vascular occlusion clamps.
- **Renal artery bleeding:** compress the **aorta above the renal artery** → clamp the arterial stump with a vascular clamp → repair with **two layers of running vascular suture**.
- **IVC bleeding** (avulsed or lacerated renal vein; avulsed gonadal or lumbar vein): **finger on the hole** → **grasp with an Allis clamp** → **pull up** to stop bleeding and display the defect for repair.

VASCULAR REPAIR SUTURES — POLYPROPYLENE (PROLENE), 30 OR 36 INCH (75 OR 90 CM), DOUBLE-ARMED, TAPERED NEEDLES

Vessel	Gauge	Needle
IVC or aorta	3-0 or 4-0	—
Renal vessels	4-0 or 5-0	—
Artery	—	$\frac{3}{8}$ -circle BB (17 mm) — less likely to fracture calcific plaque
Vein	—	$\frac{1}{2}$ -circle RB-1 (17 mm) or SH (26 mm)

Concept — *radical nephrectomy is governed by the order of vascular control: artery before vein, so that the kidney decompresses and the vein can be handled flaccid and inspected for thrombus — and when the artery cannot be reached laterally, it is taken at its interaortocaval origin.*

Concurrent Adrenalectomy for Renal Cell Carcinoma

- **Consider ipsilateral adrenalectomy for large upper-pole tumours** where the kidney–adrenal plane may be compromised; it is **indicated for diffuse tumour involvement or an adrenal mass on imaging**.
- **Otherwise not routine: overall adrenal metastasis is under 5%.**
- **CT and MRI miss 20–25% of adrenal metastases**, so clinical indicators of adrenal involvement must guide practice.
- **Contralateral adrenal metastasis on follow-up is nearly as common as ipsilateral** when both glands are spared — so **prophylactic ipsilateral adrenalectomy exposes the patient to adrenal insufficiency without clear benefit.**

Regional Lymphadenectomy for Renal Cell Carcinoma

- **Controversial; not required in every radical nephrectomy — overall nodal involvement is about 5%.**
- **Retrospective studies suggested benefit in carefully selected patients.** The **European Organisation for Research and Treatment of Cancer (EORTC) randomised trial** (772 patients, lymphadenectomy versus none) showed **no overall survival benefit**, but enrolled many small, low-stage tumours unlikely to benefit.
- **Later large cohorts found no survival benefit from staging dissection in the cytoreductive setting or in high-risk localised disease; a meta-analysis found no therapeutic benefit** → the field has moved away from prophylactic staging dissection for most patients.
- **Lymph node dissection does not alter cancer-specific survival, but may improve retroperitoneal recurrence-free survival and definitely provides prognostic information** → **recommended when nodes are enlarged on imaging**, for diagnosis and prognosis.
- **Probability of node positivity rises with size: maximum short-axis diameter of 1.5 cm or more ≈ 50% chance of positive nodes** (Fig. 141.25: a sigmoid risk curve against short-axis diameter).
- **Some surgeons perform staging dissection even without clinical lymphadenopathy when node positivity is likely:** cytoreductive surgery for metastatic disease, **tumour over 10 cm, nuclear grade 3 or higher, sarcomatoid histology, tumour necrosis on imaging**, extrarenal extension, tumour thrombus, and direct invasion of adjacent organs.

LYMPHADENECTOMY TEMPLATES: RIGHT VERSUS LEFT (FIGS. 141.26, 141.28)

	Right-sided mass	Left-sided mass
Stations	Paracaval, precaval, retrocaval, interaortocaval	Para-aortic (anterior, lateral), retroaortic; interaortocaval only if palpable, imaged or extensive periaortic disease
Cranial limit	Right crus (3–4 cm above the right renal vein); lymphatic tissue split to the superior border of the LRV	Level of the LRV

	Right-sided mass	Left-sided mass
Caudal limit	IVC bifurcation; aortocaval packet to the common iliac vessels	Past the inferior mesenteric artery (IMA) to the aortic bifurcation
Key vessels	Right gonadal vein ligated at the IVC with 2-0 silk (avoid avulsion); lumbar veins (typically 4–5 per side) ligated with 3-0 silk; IVC elevated with a vein retractor, then rolled medially with two sponge sticks	Lumbar arteries secured as the aorta is elevated; aorta rolled medially; SMA and coeliac trunk preserved; IMA may be divided if nodes are involved
Critical step	Clip the lymphatic trunks above the renal vein — chyle from the cisterna chyli and thoracic duct otherwise causes chylous ascites	Tissue on the anteromedial aorta clipped, divided and rolled laterally

Wound Closure

- **Check haemostasis and the adjacent organs.** Diaphragm and pleura may be injured by retraction.
- **Pleural test:** fill the retroperitoneum with saline to the level of the flank incision → anaesthetist gives **high inspiratory volumes** → **bubbling indicates pneumothorax**.
- **Small pleural injury:** running nonabsorbable suture → before completion, tip of a **14-Fr red rubber catheter** in the pleural cavity, other end under saline → deep inspiratory breath expels air → when bubbling ceases, withdraw the catheter and **cinch the closure airtight** → **postoperative chest radiograph in every case**, even when pneumothorax is not suspected.
- **Fascia in two layers:** transversus and internal oblique fasciae together; external oblique separately.
- **Wound infiltration:** 1:1 bupivacaine 0.5% and lidocaine 1%. Subcutaneous tissue **3-0 absorbable**; skin staples or subcuticular 4-0 poliglecaprone 25 (Monocryl).

Vena Caval Tumour Thrombectomy

- Many retroperitoneal tumours produce venous thrombus. Children: **Wilms tumour**, clear cell sarcoma of the kidney, adrenocortical carcinoma, neuroblastoma. Adults: **urothelial carcinoma of the renal pelvis**, lymphoma, retroperitoneal sarcoma, adrenocortical carcinoma, phaeochromocytoma, angiomyolipoma.
- **RCC is the commonest cause of IVC tumour thrombus; 5–10% of RCCs have venous tumour thrombus.**
- **Untreated IVC thrombus carries high mortality — most patients die within a year — so surgery, before or after systemic therapy in the metastatic setting, is the backbone of treatment.**
- **Venous thrombus marks aggressive disease:** regional nodal involvement, metastases and perirenal fat invasion are all common.
- **Thrombectomy is usually combined with radical nephrectomy and regional lymph node dissection**, which also helps expose the IVC and renal artery origin.

Tumour Thrombus Level and Bland Thrombus

- **Thrombus is often an unexpected CT finding with urgent referral;** patients are frequently already anticoagulated.
- **Best imaging: magnetic resonance venography** — better than most CT for **cephalad and caudal extent**, **IVC wall invasion and bland thrombus**; MRI is independent of contrast timing and so the more reliable measure of extent, though **well-timed reconstructed CT venograms** are also excellent.
- **Two components: tumour thrombus** (tumour cells contained within thrombus) versus **bland thrombus** (coagulum without tumour cells). Venous obstruction promotes bland thrombus. **Distinguishing them is the basis of operative management.**

TABLE 141.1 TRADITIONAL STAGING AND MANAGEMENT OF INFERIOR VENA CAVAL THROMBI (MAYO CLINIC LEVELS; DATA FROM BLUTE ET AL., 2004)

Level	Cranial extent	Management of tumour thrombus
0	Renal vein only	Radical nephrectomy
I	IVC, less than 2 cm above the renal vein ostium	IVC milking, partial IVC occlusion , ostial cavotomy
II	IVC more than 2 cm above the ostium, below the hepatic veins	Complete IVC mobilisation and control , infrahepatic cavotomy
III	At or above the (major) hepatic veins, below the diaphragm	Complete occlusion : suprahepatic IVC clamping + infrahepatic cavotomy. Partial occlusion : venovenous bypass + infrahepatic cavotomy
IV	Above the diaphragm	Deep hypothermic arrest , infrahepatic cavotomy, right atriotomy

- **What the table is for:** each level adds one tier of vascular control — renal vein, partial caval clamp, full caval control, hepatic inflow control, then the heart (Fig. 141.29).
- **The pattern:** level 0 thrombi are by far the commonest; the supradiaphragmatic minority consume most of the resources.
- **The traps:** level III is defined by the **true (major) hepatic veins, not accessory or short hepatic veins**; and within level III it is the **completely occlusive** thrombus that tolerates suprahepatic clamping (collaterals), whereas the **partially occlusive** thrombus needs bypass.

TABLE 141.2 THE MAYO CLINIC THROMBUS GROUPING SYSTEM FOR INFERIOR VENA CAVAL THROMBI (DATA FROM BLUTE ET AL., 2004)

Group	Associated bland thrombus	Additional IVC management
A	None	None
B	At or below the common iliac veins	Infrarenal IVC filter (e.g., Greenfield)
C	Infrarenal IVC, separate from tumour thrombus	Infrarenal IVC interruption with vena caval clip
D	Infrarenal IVC, mixed with tumour thrombus	Infrarenal IVC resection

- **What the table is for:** it **complements the levels** by adding the location and extent of bland thrombus and its consequence for the infrarenal IVC (Fig. 141.30).
- **The pattern:** group A is the great majority; management escalates filter → interruption → resection as bland thrombus moves from the pelvis into the cava and finally mingles with tumour.
- **The trap:** separate bland thrombus is interrupted, not extracted; only admixture with tumour mandates resection.

Preoperative Considerations

- *Pulmonary embolism, anticoagulation and IVC filters.* Renal tumours raise pulmonary embolism risk (malignant hypercoagulability plus thrombus embolism) → **screen for concurrent pulmonary embolism and lower-limb DVT (Doppler ultrasound)**.
- **Therapeutic anticoagulation is recommended for:** concurrent VTE, existing bland thrombus, and current or impending IVC occlusion by bulky thrombus — and should be considered even without these if evaluation will be delayed. Evidence is limited; potential benefits are **fewer pulmonary emboli, tumour thrombus shrinkage, and shrinkage or prevention of bland thrombus**.
- **Temporary suprarenal IVC filters are not used:** risk of **contralateral renal and hepatic vein thrombosis**, of **provoking embolisation**, and of **obstructing later thrombectomy** — thrombus grows onto the device and invades the IVC, enlarging the reconstruction required.

- **Transoesophageal echocardiography (TEE) is recommended for level II–IV thrombi** because of intraoperative detachment and interval growth just before surgery.
- **Angioembolisation for thrombus is seldom used:** ischaemic flank pain and **little reduction in operative difficulty**. The historical rationale — **one-third of tumour thrombi have an independent arterial supply** from the renal artery and/or aorta, so infarction might shrink a thrombus and avoid bypass or extensive liver mobilisation — is rarely borne out.
- **It may be considered when thrombus invades intrahepatic or suprahepatic veins and cannot be excised because of Budd–Chiari syndrome, or when the tumour or thrombus is associated with a bleeding kidney** preoperatively or on systemic therapy. Timing is uncertain, usually **within 1 day before surgery**; the risk of iatrogenic tumour embolism during angiography appears minimal.

SPECIALIST INVOLVEMENT BY THROMBUS LEVEL

Level	Specialist	Purpose
II–III	Vascular and hepatobiliary surgeon (if the urologist does not routinely handle the IVC and aorta)	Liver mobilisation, caval control, reconstruction
III–IV	Cardiothoracic surgeon — essential, planned preoperatively	Mediastinal access for bypass and thrombus removal
III–IV	Cardiologist or cardiac anaesthetist	Intraoperative TEE

Level I Vena Caval Thrombectomy: Right-Sided Tumour

- **Level I thrombi are usually partially occlusive and nonadherent, needing neither extensive IVC dissection nor bypass — but some invade the caval wall** (Fig. 141.31).
- **Kidney-last (less embolism) or kidney-first strategies both exist; a friable thrombus or one that has already shed pulmonary emboli → IVC control before significant tumour manipulation.**
- **Sequence:** midline or anterior subcostal access → great vessels and hilum exposed **with minimal manipulation of renal vein and IVC → right renal artery secured interaortocavally with 0 silk or a large clip** (early ligation cuts renal flow and blood loss) → kidney mobilised **outside the renal fascia**; IVC dissected above the right renal vein → **LRV, suprarenal IVC and infrarenal IVC encircled with vessel loops → Rummel tourniquets** (3–6-inch lengths of **18-Fr red rubber catheter** threaded on the vessel loop) or vascular clamps; full control is prudent before any IVC entry → **starting cranially, the IVC is pinched closed, then tourniquets applied in order: infrarenal IVC → LRV → suprarenal IVC → IVC milked toward the right renal ostium → C-shaped Satinsky clamp around the ostium, partially occluding the IVC, thrombus confirmed within the jaws before full closure** (Fig. 141.33) → IVC palpated for other thrombus → suction, two sponge sticks and laparotomy sponges placed around the vein to catch spillage → **ostium incised circumferentially** (scalpel, fine Metzenbaum or Potts scissors), **preserving at least half the IVC width for closure** → thrombus extracted by **gentle downward traction on the renal vein and ring forceps** → **gauze wrapped over the renal vein stump and tied with silk** against spillage (Fig. 141.34) → medial attachments divided, **renal artery ligated again before division**.
- **Closure (Fig. 141.35):** IVC inspected for residual thrombus, lumen irrigated with heparinised saline (**100 U/mL**) → **running 4-0 Prolene on a BB needle** → before tying: **release infrarenal IVC and LRV clamps and evacuate clot** → anaesthetist applies **positive airway pressure and holds ventilation** → **infrarenal IVC pinched, suprarenal Satinsky released** → **5–10 mL of blood allowed out of the defect** to flush air, thrombus fragments and debris → suture tied → haemostasis, irrigation; **no drain needed**.

Level II Vena Caval Thrombectomy: Left-Sided Tumour

- **Harder, because the IVC is best reached from the right: both colons must be mobilised. Midline or chevron incisions give the best access** (Fig. 141.36: level II thrombus in a completely duplicated IVC, with patch repair and rerouting of the left IVC).
- **Left side first:** left colon mobilised, left anterior pararenal space developed → **left renal artery ligated near its aortic origin** → **confirm arterial control by palpation or Doppler probe** — a persisting pulse means an **accessory artery** or incomplete control (e.g., calcification) → **adrenal, lumbar and gonadal branches of the LRV ligated and divided** (often dilated, friable, occasionally thrombosed) → kidney mobilised outside the renal fascia, **left adrenal left in situ**, left ureter divided with a tie on the stay side.
- **Right side next:** right colon and small bowel mobilised → **Kocher manoeuvre** → right anterior space developed, great vessels exposed → IVC dissected to its bifurcation, **right gonadal vein ligated** on its anterior surface → **split-and-roll manoeuvre** frees the IVC above and below the thrombus with lumbar vein ligation → optionally **ligate accessory (short) hepatic veins to release the caudate lobe, gaining 2–3 cm of infrahepatic IVC** → **sequential control: (1) infrarenal IVC, (2) contralateral (right) renal vein, (3) suprarenal IVC.**
- **Contralateral renal artery:** historically clamped to prevent engorgement, which matters mainly for left tumours because the **right kidney lacks venous collaterals**. The authors **no longer clamp the right renal artery**; engorgement is relieved by **periodically releasing the right renal vein clamp** — preferable to right renal ischaemia.
- **Heparin:** some give **0.5 mg/kg IV** before clamping; the authors find **bleeding, not clotting, is the principal problem** and do not heparinise routinely.
- **Thrombectomy (Fig. 141.38):** LRV ostium circumferentially excised, incision extended superiorly onto the anterolateral IVC with **Potts scissors** → **Penfield dissector** extracts the thrombus → troublesome **lumbar veins ligated or sutured**; sponge sticks isolate and compress bleeding segments → **thrombus and kidney removed en bloc** → IVC flushed with heparinised saline, **intima inspected; suspicious areas biopsied or resected** → during prolonged clamping, **periodically release the contralateral renal vein and inferior IVC clamps** to clear coagulum.
- **Closure:** the IVC lumen may be narrowed safely to about 50% of its preoperative size; greater narrowing (residual lumen under 50%) → **bovine pericardial patch, smooth side luminal, rough side outward**, length equal to the defect, **width by the Leibovich–Bower equation** (see Patch Cavoplasty) → **running 4-0 Prolene** → before tying, **infrarenal clamp released and 5–10 mL of blood let out** to clear air and debris → after tying, **contralateral renal clamp released, then the suprarenal IVC clamp** → regional lymphadenectomy; closed suction drain considered; irrigation and closure.

Level III Vena Caval Thrombectomy: Intra-Abdominal Approach

- **Intrahepatic thrombi are very challenging.** Theatre set up for possible **cardiopulmonary bypass (CPB)**, including **deep hypothermic arrest**; performed in vascular or cardiac theatres with their anaesthesia teams.
- **Intraoperative TEE** measures the cranial extent, **monitors thrombus fracture and embolisation**, and assesses cardiac function for haemodynamic management (Fig. 141.39: thrombus in IVC and right atrium).
- **The key decision is made intraoperatively only** — after renal artery ligation, liver mobilisation and IVC assessment: **intra-abdominal extraction with complete hepatic mobilisation, or combined intrathoracic–intra-abdominal approach with bypass.**
- **Clamping below the hepatic veins is preferred** because hepatic venous return is large.
- **Rule of thumb:** free-floating, nonocclusive thrombi tolerate suprahepatic clamping poorly and more often need bypass; completely occlusive thrombi have developed collateral venous networks and

tolerate clamping much better (Fig. 141.40: complete IVC occlusion with collateralisation to the azygos system on CT and angiography).

- **A level IV thrombus can occasionally be milked into the abdomen** through a small diaphragmatic incision and treated intra-abdominally.
- **IVC control must never compromise the operation:** bleeding and hypotension lead to **incomplete tumour resection, which is universally fatal**.
- **Incision: anterior midline preferred;** chevron with sternotomy is an alternative. Right kidney and great vessels exposed as for level I; **right renal artery ligated interaortocavally;** infrahepatic IVC dissected; **infrarenal IVC and LRV encircled** with Rummel tourniquets or clamps.
- **Liver mobilisation (liver-transplant technique; Figs. 141.41–141.42):** **ligamentum teres** (obliterated left umbilical vein, at the free border of the falciform) ligated and divided → **falciform ligament** divided with cautery to the upper liver border, where it branches into the **coronary ligament (right) and left triangular ligament** → **superior layer of the coronary ligament** divided, sparing the liver and the IVC behind it in the **bare area** → continued along the right border to the **right triangular ligament** (fused superior and inferior coronary layers), divided → **inferior layer of the coronary ligament** (liver to diaphragm) divided up toward the IVC → **left triangular ligament** divided anteriorly, then posteriorly toward the IVC → **right lobe rotated to the midline**, exposing the IVC on the posterior liver. A hepatobiliary surgeon is valuable here.
- **Left-sided tumours:** divide the diaphragmatic attachments of the spleen so that it rotates medially with the pancreas without trauma (**partial Mattox manoeuvre**).
- **Retrohepatic plane — the hepatic veins fall into two groups:**
 - **Upper group — right, middle and left hepatic veins:** the principal hepatic outflow, never divided. Thrombus may extend into them and must be cleared. **Their obstruction produces Budd–Chiari syndrome, a contraindication to proceeding with radical nephrectomy and thrombectomy.**
 - **Lower group — accessory hepatic veins** (draining mainly the caudate lobe, a little of the right lobe): **safely divided**, ligated with 3-0 silk on IVC and liver sides.
- **Lumbar veins ligated with 2-0 silk;** the plane behind the IVC developed → **IVC fully mobilised**.
- **Pringle manoeuvre:** if the thrombus does not descend after arterial ligation to permit clamping below the main hepatic veins, **portal inflow must be controlled** → window in the **lesser omentum** → **porta hepatis** (portal triad; hepatic pedicle — portal vein, common hepatic artery, common bile duct) encircled with a Rummel tourniquet. **Clamping it prevents hepatic engorgement and massive blood loss when the IVC is clamped above the major hepatic veins.**
- **Total hepatic vascular occlusion = IVC clamped above and below the hepatic veins plus Pringle.** If the IVC is clamped **below** the major hepatic veins with accessory veins ligated, Pringle may be unnecessary.
- **Normothermic Pringle limit: up to 60 minutes, but 20 minutes or less preferred** — ischaemic liver injury and **portal vein thrombosis;** also **splenic engorgement and rupture** from backed-up splenic venous drainage.
- **Resectability is judged by TEE and intraoperative anatomy:** thrombus below, or milkable below, the hepatic veins → usually no bypass; thrombus involving the hepatic veins or above the liver → bypass may be needed (Fig. 141.44: thrombus to the diaphragm and into the right atrium, eventually requiring CPB with deep hypothermic arrest).
- **Clamping trial:** occlude the IVC above the liver and thrombus; observe for **2–5 minutes**.

HAEMODYNAMIC EFFECT OF SUPRAHEPATIC IVC CLAMPING

Parameter	Change
Cardiac preload	↓ 60%

Parameter	Change
Peripheral vascular resistance	↑ 80%
Heart rate	↑ 50%
Cardiac output	↓ 40%
Mean arterial pressure	↓ 10–20%
Not tolerated if	Cardiac output falls more than 50% or mean arterial pressure falls more than 30% → bypass (preferred) or supraceliac aortic clamping
Proceed intra-abdominally if	Trial tolerated and thrombus removable in under 30 minutes

- **Sequence:** clamp infrarenal IVC → contralateral (left) renal vein → porta hepatis → suprahepatic IVC → right renal vein ostium incised circumferentially, incision extended up toward the intrahepatic IVC, large enough for complete extraction and **intimal inspection** → thrombus and kidney excised (Fig. 141.45) → adherent thrombus cleared with a **Penfield dissector**; a **Fogarty balloon** or **20-Fr Foley catheter** serves as embolectomy catheter for thrombus out of reach → wall involved by tumour that cannot be scraped away → **partial or complete IVC resection and reconstruction** → during deep hypothermic arrest, a **cystoscope** can inspect hepatic veins and suprahepatic IVC through a smaller cavotomy.
- **Minimising Pringle time:** isolate the IVC circumferentially above the hepatic veins (initial clamps) and below them → after extraction, **place an infrahepatic clamp, release the suprahepatic clamp, then release the Pringle** → close as for level II → **tack the hepatic ligaments back to prevent liver torsion.**

Level III–IV Vena Caval Thrombectomy: Combined Intra-Abdominal and Intrathoracic Approach

- **For level III thrombi not removable intra-abdominally and nearly all level IV thrombi** (Fig. 141.46).
- **Incisions:** thoracoabdominal, chevron laparotomy with sternotomy, or anterior midline laparotomy with sternotomy (preferred). A cardiothoracic surgeon participates.
- **Sequence:** abdominal phase as for the intra-abdominal approach → **cardiothoracic surgeon called after releasing the caudate lobe but before rotating the liver** (mobilisation is easier after sternotomy) → **median sternotomy, pericardium opened, right heart exposed** (Fig. 141.48) → bypass instituted → renal vein ostium excised, incision extended cranially on the IVC, thrombus extracted; **right atriotomy** usually needed for suprahepatic thrombus (Fig. 141.52) → atrium and IVC closed → off bypass, **chest tubes placed** → hepatic ligaments tacked back.

Bypass Techniques for Inferior Vena Caval Surgery

- **Bypass complicates and prolongs thrombectomy but is often essential** for a safe and complete operation.
- **Consider bypass when:** the IVC cross-clamping trial causes significant hypotension; there is **preoperative cardiac or hepatic dysfunction, contralateral renal dysfunction or portal hypertension**; or there is **major intraoperative bleeding** that is difficult to control.

BYPASS TECHNIQUES COMPARED

	Venovenous bypass (VVB)	CPB without arrest	CPB with deep hypothermic circulatory arrest (DHCA)
Principle	Pump shunts infrarenal venous blood to the heart	Full extracorporeal circulation	Heart stopped, patient cooled, blood drained
Chest opened	No (percutaneous) — key advantage	Yes	Yes
Heparinisation	Not full — less postoperative bleeding	Full	Full

	Venovenous bypass (VVB)	CPB without arrest	CPB with deep hypothermic circulatory arrest (DHCA)
Main drawback	Intercostal and lumbar flow continues → major bleeding at cavotomy	Coagulopathy	Invasive; coagulopathy; cardiac and neurological risk
Avoid when	Atrial thrombus not milkable into the IVC; extensive bland thrombus in iliac veins or infrarenal IVC; pre-existing Budd-Chiari syndrome	—	—
Time for thrombectomy	—	—	Up to 60 minutes (under 40 better) versus ~30 minutes for clamping without bypass
Authors' preference	Least invasive option	—	Preferred for level IV

- **Venovenous bypass. Inflow cannulation:** percutaneous **femoral vein**, or direct intraoperative **IVC just above its bifurcation**. **Return:** percutaneous **internal jugular vein**, **brachial/axillary vein cutdown**, or directly into the **right atrium**. **Place the cannula tip as far from the thrombus as possible** — dislodgement causes massive pulmonary embolism, and tumour cells may be aspirated and recirculated.
- **Percutaneous VVB sequence (Fig. 141.49):** immediately after intubation, anaesthetist inserts an **8–18-Fr heparin-bonded cannula into the internal jugular vein (Seldinger)** → femoral vein: **6-cm 18-gauge needle**, guidewire, tract dilated, **14–20-Fr heparin-bonded cannula advanced to the common iliac vein** → both connected to the pump with heparin-bonded tubing (portal vein cannulation with 20 Fr possible, rarely needed) → incision, kidney and IVC dissected → all vessels clamped, **pump started** → thrombectomy on pump, ligating troublesome lumbar and intercostal veins → **unclamp in the same order as clamped**.
- **Open VVB sequence:** kidney and IVC dissected, liver mobilised, sternotomy → **20-Fr cannula in the infrarenal IVC well away from the thrombus**, then **14–20-Fr cannula in the right atrial auricle** → infrarenal IVC, renal vein, porta hepatis and suprarenal IVC clamped → bypass started → thrombectomy as fast as possible.
- **Cardiopulmonary bypass with and without deep hypothermic arrest. Five advantages of DHCA:** (1) usable when the thrombus **cannot be milked below an intrapericardial IVC clamp**; (2) **no need to clamp aorta or porta hepatis or ligate many lumbar and hepatic veins** — though every traumatised or transected vessel must be controlled before coming off bypass; (3) **bloodless field for complete inspection** of IVC and hepatic veins; (4) **lower embolisation risk**; (5) **longer working time**.
- **DHCA sequence (Fig. 141.51):** kidney and IVC dissected, liver mobilised → sternotomy, pericardium opened → **heparin-bonded venous cannulae in infrarenal IVC and right atrium; arterial cannula in the aortic arch** → heparinise, start bypass → **aorta clamped, cardioplegia given** → **perfusate cooled to 10–14°C; patient cooled 15–30 minutes to a core of 16–18°C**, with **electroencephalography** confirming cerebral cooling → pump stopped, **95% of blood volume drained to the reservoir** → thrombectomy as fast as possible, ligating all potential bleeders (Fig. 141.52) → coronary artery bypass may be added for known coronary disease → if prolonged: **10 mL/kg/min trickle perfusion or retrograde cerebral perfusion** → IVC and right atrium repaired → warm blood reinfused, CPB restarted → **haemostasis while rewarming to 37°C over 30–45 minutes** → heart restarts, bypass stopped, cannulae removed, **protamine given** → **coagulopathy is common: fresh frozen plasma, platelets and packed red cells ready** → thoracostomy tubes and closed suction abdominal drains.

Patching, Replacing, and Interrupting the Inferior Vena Cava**IVC RECONSTRUCTION OPTIONS COMPARED**

	Patch cavoplasty	Vena caval replacement	Filtration / permanent interruption
Indication	Residual IVC expected under 50% of original size — prevents stenosis and thrombosis	Circumferential resection, or defect too large to patch	Infrarenal bland thrombus
Material / device	Autologous or bovine pericardium, PTFE, collagen-impregnated Dacron, autologous saphenous vein	Polytetrafluoroethylene (PTFE) graft, 16–20 mm (authors); alternatives spiral saphenous vein, superficial femoral vein, tubularised pericardium	Filter (pelvic veins only); serrated caval clips (Adams–DeWeese, Moretz), vascular GIA stapling, suture plication, suture ligation
Suture	Double-armed 5-0 Prolene, BB needle	—	—

- *Patch cavoplasty* (Fig. 141.54). Patch cut **slightly larger than the defect, oval**; width by the **Leibovich–Bower equation** (Dr Brad Leibovich, urological oncologist, and Dr Thomas Bower, vascular surgeon, Mayo Clinic):

$$\text{Patch width} = \text{diameter of the normal cava} \times \% \text{ caval diameter resected} \times 3$$

- **Small, regular bites** give a tight closure; **avoid intraluminal inversion of the edges** — the rough, thrombogenic surface must stay extraluminal.
- **Two methods: tack both apices and run from each apex to the midpoint (four knots), or parachute and sew circumferentially (one knot).** Handle the patch minimally; liberal heparinised saline aids vision.
- **Before final closure:** release the **caudal IVC clamp**, then the **contralateral renal vein clamp** → **5–10 mL of venous blood** flushed out → **cranial clamp released after a Valsalva breath-hold**.
- *Vena caval replacement* (Fig. 141.55). **Large graft diameter reduces thrombosis. Sequence:** liver mobilised, IVC fully exposed → bypass if indicated; **5000 units IV heparin** → IVC clamped, affected segment excised → **cranial anastomosis first** → graft clamped, cranial IVC clamp released to **test the upper anastomosis** → graft sized and cut → **caudal anastomosis** → before tying: graft unclamped, **5–10 mL blood flushed**; graft reclamped; infrarenal IVC unclamped, **5–10 mL flushed** → final knot, IVC unclamped → **graft wrapped in omentum or retroperitoneal fat**; hepatic ligaments reapproximated against torsion → **contralateral renal vein sutured perpendicularly into the PTFE graft** after removing the rings.
- **Postoperatively:** low-dose IV heparin or reduced-dose low-molecular-weight heparin; lifelong aspirin 81 mg; full anticoagulation may be considered.
- *Filtration and permanent interruption for bland thrombus.* **Bland thrombus limited to the pelvic veins** → intraoperative infrarenal vena caval filter.
- **Diffuse infrarenal IVC bland thrombus** → **permanent interruption**, preserving the **lumbar collaterals**, the "release valve" for impaired caval flow.
- **Infrarenal bland thrombus distinct from tumour thrombus** → **interruption without resection:** attempts to clear diffuse organised bland thrombus almost always fail and often injure the vessel. **Clips and staplers are easy to apply; serrated clips also allow partial flow.**
- **Bland thrombus admixed with retrograde-growing tumour thrombus** → **segmental IVC resection with permanent interruption** — tumour cannot be reliably separated from bland thrombus. **Resect below the contralateral renal vein ostium, as close to it as possible** (avoids a turbulent, thrombogenic upper stump), and **preserve the lumbar veins in the lower stump** for collateral drainage.

"SHORTCUTS" WITH VENOUS TUMOUR THROMBI THAT ARE "LONGCUTS"

Shortcut	Risk and consequence
IVC clamping and entry without isolating all inflow	Haemorrhage and poor vision at cavotomy → shock, incomplete thrombus removal
"Side-biting" the IVC for an apparently easy level I	Insufficient caval edge mobility after excision → clamp repositioning and bleeding; clamp dislodgement
Clips on the IVC side of perforators and lumbar veins	Clips prevent clamp occlusion and compromise vision
Incomplete arterial control	Profuse bleeding from the renal vein stump; thrombus shedding at cavotomy
Excessive traction on, or milking of, thrombus to avoid further cranial dissection	Tumour embolism or IVC laceration
Clamping before anaesthesia has hydrated the patient, or skipping the test clamp	Profound hypotension with the IVC open; cardiac or end-organ ischaemia

Concept — *caval thrombectomy is a sequence of ever-higher vascular control dictated by the cranial extent of tumour and the behaviour of bland thrombus below it: isolate every inflow before opening the cava, clamp from below upward, and release from below upward so that air and debris are flushed out before the cranial clamp comes off.*

PARTIAL NEPHRECTOMY FOR MALIGNANT DISEASE

- Partial nephrectomy is the preferred operation for renal neoplasms when technically feasible and when preserving renal function is a priority (Fig. 141.56: contrast CT, 3D reconstruction of a large intrarenal component, and arterial reconstruction showing a lower-pole artery close to the mass).
- Historically reserved for imperative indications — bilateral tumours, tumour in a solitary kidney, high risk of future renal failure — and small tumours under 4 cm. Indications have broadened to **most renal masses that can be removed safely and completely, independent of size.**

RELATIVE CONTRAINDICATIONS TO PARTIAL NEPHRECTOMY

Technical	Cancer-related
Less than 20% of global nephron mass retained	Diffuse encasement of the renal pedicle by tumour
	Diffuse invasion of the central collecting system
	Tumour thrombus in major renal veins or the IVC
	Adjacent organ invasion (cT4)
	Regional lymphadenopathy (cTxN1)

Preoperative Considerations

- The preoperative considerations for radical nephrectomy apply, plus those unique to partial nephrectomy.

Hyperfiltration Injury

- **Mechanism:** removal of substantial parenchyma delivers renal blood flow to fewer nephrons → **raised glomerular capillary perfusion pressure** → **raised single-nephron GFR (hyperfiltration)** → over years to decades, **focal segmental glomerulosclerosis, proteinuria and progressive renal failure.**
- Rare after partial nephrectomy; usually seen when total nephron mass of both kidneys is reduced by more than 80%.

Renal Ischaemia and Hypothermia

- **Vascular compression limits blood loss and improves vision:** manual compression, a **parenchymal compression clamp (Kaufmann)**, **selective renal artery clamping** (with or without the vein), or **en bloc clamping of the whole pedicle** (Fig. 141.57: Satinsky clamp across the parenchyma of a large polar tumour). **Leaving the vein unclamped for retrograde perfusion has no proven benefit.**
- **Duration of ischaemia and long-term function remain debated.** Early landmark studies showed **longer ischaemia → worse function**, and **surgeons traditionally aim for under 30 minutes.**
- **Once the proportion of healthy parenchyma preserved is accounted for, ischaemia time matters less or not at all** — but all such studies lay **within conventional limits**; the effect of **prolonged warm ischaemia remains unknown.** → **Judicious, individualised ischaemia**, weighing baseline function and expected preserved parenchyma.
- **Adjuncts to limit ischaemic injury:** renal hypothermia, IV mannitol, IV diuretics.

NEPHROPROTECTIVE MEASURES IN PARTIAL NEPHRECTOMY: THE RANDOMISED EVIDENCE

Question	Trial design	Verdict	Caveat
Renal hypothermia versus none (with ischaemia)	Randomised controlled trial	No difference in postoperative renal function	Cohort enriched for contralateral kidney present and good baseline function ; role in CKD or solitary kidney unclear
IV mannitol versus placebo	Prospective placebo-controlled randomised trial	No clinical benefit	Dose lower than routine , no concurrent diuretic, excellent baseline function
On-clamp versus off-clamp (fully perfused)	Two prospective randomised trials (retrospective data mixed)	No difference in any clinical outcome	Carefully selected patients; ischaemia within accepted limits
Digital 3D anatomical models for planning	Randomised clinical trial	Shorter operative time, less blood loss, shorter ischaemia, shorter stay	—

- **Bottom line:** hypothermia and mannitol are of limited use with normal renal function and a healthy contralateral kidney; in compromised function or a solitary kidney their value is unclear, and both may be used at the surgeon's discretion.

Multifocality and Tumour Size

- **Multifocality:** about 2% in clear cell and chromophobe RCC, about 10% in papillary RCC, and more common as tumour size rises (Fig. 141.58: hilar and peripheral tumours on 3D CT).
- **Inspect the entire renal surface** at operation to confirm the imaging.
- **Partial nephrectomy remains viable for multifocal tumours** if all can be resected safely and the remnant reconstructed — including **simultaneous bilateral partial nephrectomy for synchronous bilateral tumours** in selected patients.

Hereditary Renal Malignancy

- Hereditary tumours are usually multifocal, bilateral and recurrent.
- **Except in hereditary leiomyomatosis and RCC (HLRCC)** — treated aggressively with wide excision — most hereditary syndromes can be observed safely until tumours reach 3 cm, with little risk of metastasis.
- **At partial nephrectomy:** preserve perirenal fat and renal fascia, visualise the whole surface, **resect all visible tumours**, and use intraoperative ultrasound to find subcortical tumours.

- **Reapproximate perirenal fat and renal fascia after renorrhaphy** — prevents visceral adhesion to the capsule and eases future surgery.

Digital Models of Three-Dimensional Renal Anatomy

- **Traditionally, planning relied on cross-sectional images** (hilar vessels; tumour relation to sinus, vessels and collecting system).
- **3D digital renderings from cross-sectional imaging reveal anatomy previously hard to appreciate** (Fig. 141.60: printed model — tumour, collecting system, veins, arteries); a randomised trial showed benefit (tabulated above).

Surgical Technique

Approach

- **Extraperitoneal (flank incisions) or transperitoneal (midline or subcostal)**. Surgeons should be comfortable with all and individualise by **habitus, prior surgery, hilar anatomy, tumour size and location, and comorbidity**.

Vascular Control and Renal Mobilisation

- **Achieve hilar control early**, identifying the main renal artery and vein and any accessory vessels — **even when ischaemia is not planned**, so clamps can be applied promptly if needed.
- **Anticipate anomalies: retroaortic renal vein, precaval renal artery, early or multiple branching, accessory vessels**.
- **Mobilisation: at minimum, the capsule around the tumour exposed circumferentially** to delineate borders and permit capsular renorrhaphy; full mobilisation helps deliver the tumour or excise multiple tumours.
- **Reapproximate Gerota's fascia at the end** to prevent small bowel adhesion to the capsule.

Renal Perfusion During Partial Nephrectomy

- **Traditional technique clamps artery and vein**. Modifications: **omission of venous clamping; selective clamping of tumour-supplying branches; early unclamping** after excision or the first renorrhaphy layer; **fully perfused ("off-clamp") surgery**.
- **Verdict (randomised evidence tabulated above): ischaemia within widely accepted limits is safe in selected patients; on-clamp versus off-clamp remains the surgeon's individualised choice**.

Tumour Excision Techniques

ENUCLEATION VERSUS RESECTION (FIG. 141.61)

	Tumour enucleation	Tumour (wedge) resection
Plane	Along the tumour pseudocapsule — between normal parenchyma and pseudocapsule	Through normal parenchyma, 5–10 mm peripheral to the tumour
Normal tissue removed	Minimal	Rim of normal parenchyma on the specimen
Occlusion	May be omitted for small lesions ; manual compression or hilar clamping if bleeding obscures the margin	Usually required

	Tumour enucleation	Tumour (wedge) resection
Bleeding	Often minimal if the plane is followed circumferentially	Greater risk of transecting small vessels → meticulous haemostasis at renorrhaphy

- **Enucleation sequence:** cortex scored circumferentially with cautery → sharp and blunt dissection in the pseudocapsular plane → small intrarenal vessels **ligated, then divided sharply** → tumour excised, margins inspected for gross positivity; **deep margin sent for frozen section** as needed.

Intraoperative Ultrasound

- **Exophytic tumours** are excised along the visible contour; **endophytic or centrally extending tumours** need ultrasound to localise borders.
- **Performed by the surgeon after defatting kidney and tumour, before hilar clamping or excision;** formal intraoperative radiology when borders remain unclear.
- **Goals:** tumour location and borders, **depth of parenchymal penetration**, relation to vessels and collecting system, and planning of excision (Fig. 141.62: endophytic hilar tumour clearly distinct from parenchyma).

Renorrhaphy Techniques

TWO-LAYER RENORRHAPHY

	Inner layer	Outer layer
Target	Vessels and collecting system in the resection bed	Capsule and parenchyma over the defect
Technique	Small vessels: 4-0 absorbable figure-of-eight on a tapered needle , or argon beam or bipolar coagulation. Collecting system checked and closed with running absorbable suture (Fig. 141.63)	Running or interrupted absorbable suture (authors: interrupted, tapered needle): capsule → bed at the corticomedullary junction → back through it → capsule on the far side
Adjuncts	—	Haemostatic bolster (Fibrillar or Nu-Knit) in the bed before tying
Cautions	Avoid incorporating adjacent vessels (delayed bleeding as haematuria); no permanent sutures — stone nidus	Tension parallel to the suture, never upward traction — soft parenchyma tears

- **Completion:** clamps released → renorrhaphy inspected immediately → **manual compression**, then additional sutures, until haemostatic → perirenal fat and renal fascia reapproximated → **collecting system entered** → **closed suction drain in the pararenal space** to monitor for urine leak.

Concept — *partial nephrectomy is a balance between two losses — nephrons removed with the margin and nephrons injured by ischaemia — and the evidence now says the quantity of parenchyma preserved outweighs minutes of clamp time within accepted limits.*

INTRAOPERATIVE AND POSTOPERATIVE COMPLICATIONS

Intra-Abdominal

Injury to the Intestinal Vasculature

MESENTERIC VESSELS AT RISK: CONSEQUENCE AND MANAGEMENT (FIG. 141.64)

Vessel	Supplies / drains	Mechanism of injury	Consequence	Management
IMA and branches	Left colic → distal transverse and descending colon; sigmoid branches → sigmoid	Reflection of the left mesocolon off Gerota's fascia; deliberate division for colonic invasion	Usually none — collateral supply from the SMA via the marginal artery of Drummond	Ligation safe if the marginal artery is patent
SMA	Entire small intestine, caecum, right colon, proximal transverse colon	Mistaken for the left renal artery in left nephrectomy; traction injury during Cattell-Braasch; retractor blade	Potentially lethal	Reverse immediately: vascular surgeon; primary repair or graft bypass
IMV	Descending mesocolon, just lateral to the ligament of Treitz	—	None	Safely ligated; landmark — incise posterior peritoneum just medial to it
SMV	Root of mesentery; joins splenic vein to form the portal vein	—	Bowel oedema, venous engorgement, impaired portal return → systemic hypotension/splanchnic hypertension syndrome (venous thrombosis, bowel ischaemia, necrosis)	Never ligate unless the only option. Laceration: clip small tributaries, isolate with atraumatic clamps, 6-0 Prolene venorrhaphy. Transection: primary reanastomosis or autologous vein graft; Gore-Tex only if no vein (high thrombosis). Do not close the abdomen primarily — abdominal compartment syndrome

- **Why the SMA is mistaken for the left renal artery:** it arises from the anterior aorta about 1–2 cm above the left renal artery and deflects slightly leftward.
- **Why Cattell-Braasch endangers the SMA:** the mobilised small bowel, mesentery and right mesocolon pivot on the SMA, which itself leaves the aorta anteriorly at a wide angle; excessive anterior displacement or a retractor blade against it causes injury.
- **Fig. 141.64:** the ascending mesocolon is supplied mainly by the **right colic artery**, the descending mesocolon mainly by the **left colic artery**; the **marginal artery of Drummond** links SMA, IMA and rectal arteries, sustaining the left colon and rectum.

Injury to the Liver and Spleen, Duodenum, Pancreas, and Pleura**VISCERAL INJURIES: MECHANISM AND MANAGEMENT**

Organ	Mechanism	Recognised intraoperatively	Late or unrecognised
Liver and spleen	Traumatic retraction during mobilisation	Capsular tears, minor lacerations: argon beam or cautery (spray, setting 100) ; fibrin glue, oxidised cellulose (Surgicel). Serious splenic injury → splenorrhaphy or splenectomy. Minor hepatic laceration: absorbable suture on ½-circle tapered needle with Nu-Knit pledgets	—
Duodenum	Right-sided surgery (Kocherisation) ; Cattell-Braasch (transverse duodenum displaced)	Intramural haematoma: expectant; if large and narrowing the lumen → incise serosa and muscularis (not mucosa), drain, close with interrupted 3-0 silk. Minor cautery injury or laceration: debride; two layers — mucosa continuous 4-0 chromic or Vicryl; seromuscular interrupted 3-0 silk; omental flap and closed suction drain	Untreated narrowing haematoma → obstruction ; nasogastric decompression until bowel function returns
Pancreas	Left-sided surgery: reflecting mesocolon and pancreatic tail; stapling the hilum	Superficial: fibrin glue + closed suction drain ; drain monitored for alkaline pH, amylase and lipase (fistula). Deep or ductal injury → gastrointestinal surgeon	Pain, vomiting, fever, distension, ileus; raised serum amylase/lipase ; fluid collection in the left renal bed (enhancing → abscess); rarely fistula through the wound → percutaneous drainage , drain fluid amylase/lipase, antibiotics if infected, nil by mouth and nasogastric tube, octreotide for persistent output, fluid and electrolyte balance; general surgery if nonhealing
Pleura	Upper-pole mobilisation (complex or redo); left: dividing the splenorenal ligament; flank incisions (pleura drapes beneath the ribs)	14-Fr red rubber catheter into the pleural cavity on gentle suction; small defect → purse-string 3-0 chromic around it; positive-pressure breath, withdraw catheter as the suture is tied. Large defect → chest tube. Postoperative chest radiograph	Dyspnoea, chest pain, respiratory distress (pneumothorax or haemothorax) → radiograph, oxygen, chest tube

- **Duodenal trap:** a segment may look nonviable after haematoma drainage — **do not resect; the impression is false.** A general or gastrointestinal surgeon is valuable.
- **Two pleural repair methods appear in the chapter:** at wound closure, a running nonabsorbable suture with the catheter under saline; in the complications section, a 3-0 chromic purse-string with the catheter on gentle suction. Both evacuate air on a positive-pressure breath before the final tie.

Complications Specific to Inferior Vena Cava Tumour Thrombectomy

IVC THROMBECTOMY COMPLICATIONS: PREVENTION AND MANAGEMENT

Complication	Cause	Prevention / management
Air embolism (right heart, pulmonary arteries) — potentially lethal	Air entering at cavotomy	Release the caudal IVC clamp first ; let air and about 5 mL of blood escape from the repair before removing the cranial clamp
Acute pulmonary embolism (tumour or bland)	Manipulation before control	Minimal manipulation before vascular control ; early renal artery ligation; Greenfield filter for pelvic bland thrombus . Intraoperative respiratory distress → prompt thoracotomy, pulmonary arteriotomy and extraction
Massive haemorrhage	Parasitic vessels; IVC tears; bleeding despite "complete" control = an unligated lumbar vein	Ligate all lumbar veins before cavotomy . Uncontrolled bleeding off bypass → supraceliac aortic clamping or deep hypothermic CPB (itself coagulopathic); liberal FFP, platelets, red cells. Cell Saver not recommended — disseminates tumour cells
Hepatic dysfunction (raised transaminases, alkaline phosphatase)	Common after level III–IV with suprahepatic clamping and/or bypass	Minimise porta hepatis clamping ; enzymes peak at 2–3 days , then resolve
Cardiac (from CPB/arrest)	Coronary hypoperfusion; electrolyte disturbance; reperfusion	Ischaemia and myocardial infarction ; atrial and ventricular tachyarrhythmias ; transient dysfunction with heart failure (peripheral and pulmonary oedema)
Neurological (from CPB/arrest)	Cerebral hypoperfusion; electrolyte disturbance	Transient or permanent deficits ; seizures ; postoperative cognitive dysfunction and delirium in the elderly with pre-existing impairment

- **Mitigation of CPB complications:** optimal haemodynamics and **cerebral perfusion pressure**; **cerebral oximetry**; **anticoagulation of the circuit with heparin** (patency, no embolic clot), **reversed with protamine** after bypass; monitor **activated partial thromboplastin time and activated clotting time**.
- **Preoperative risk assessment for cardiac and neurological complications** allows preparation and intraoperative monitoring.

Complications Specific to Partial Nephrectomy

PARTIAL NEPHRECTOMY COMPLICATIONS COMPARED

	Urine leak	Postoperative bleeding	Arteriovenous fistula	Renal insufficiency
Cause	Collecting system entry (large, endophytic, hilar tumours), even after closure	Delayed interruption of subsegmental branches ; pseudoaneurysm ; less often AVF	Inadvertent injury creating an artery-to-vein communication (high flow, low resistance)	Solitary kidney: large tumour, excessive parenchymal removal, prolonged ischaemia ; also obstruction, drug toxicity, vascular thrombosis or disruption
Presentation	Most within 72 hours ; high drain output	Flank pain, haematuria , haemodynamic instability (rupture)	Thrill or pulsatile mass, bruit, haematuria, heart failure, renal ischaemia from steal	Usually mild and temporary ; hyperfiltration → gradual decline with proteinuria

	Urine leak	Postoperative bleeding	Arteriovenous fistula	Renal insufficiency
Diagnosis	Drain fluid creatinine above serum; IV indigo carmine if uncertain; CT urography only if symptomatic	Diagnostic angiography preferred over CT angiography	Doppler ultrasound, CT angiography, magnetic resonance angiography	—
Treatment	Drain → IR drain → retrograde double-J stent + Foley	Conservative → selective embolisation → re-exploration	Observation → coil embolisation or stent → surgery	Supportive; some need haemodialysis

- **Urine leak.** Close large collecting system defects primarily with absorbable suture. Leave a closed suction drain after open renal surgery, especially after deep resection.
- **Drain rules:** output low for 48 hours → leak unlikely → remove the drain by 48 hours. Fluid creatinine is measured only if output is high — typically over 50 mL per 24 hours.
- **High output with raised fluid creatinine but no significant symptoms** → no cross-sectional imaging needed. Symptoms — persistent abdominal pain, fever, ileus, leakage from the incision, disproportionate rise in SCr — → CT urography if renal function permits (delayed phase shows extravasation and suggests infection or abscess).
- **Treatment sequence:** maintain the closed suction drain in the urinoma → percutaneous drain by interventional radiology if drainage is inadequate → urinoma unresolved after 48 hours, or symptoms persisting beyond 24 hours → retrograde double-J stent plus Foley catheter (antegrade flow); any retrograde pyelogram gentle and low-pressure so as not to enlarge the defect → most fistulas resolve in 4–6 weeks; reoperation is rarely required.
- **Postoperative bleeding. Pseudoaneurysm:** persistent communication between a disrupted arterial wall and the surrounding tissue, forming a contained haematoma — from injury at excision, suturing or thermal energy.
- **Risk rises with tumour complexity (large, endophytic) and postoperative anticoagulation.**
- **Mild bleeding in a stable patient:** imaging not necessarily required; bed rest, hydration, close monitoring and serial blood counts.
- **Significant bleeding** (falling haemoglobin, transfusion, instability) → angiography by interventional radiology — CT angiography has more false negatives, particularly for pseudoaneurysm, and cannot treat → selective embolisation of segmental or subsegmental arteries salvages the kidney.
- **Life-threatening haemorrhage** (hypotension refractory to blood-product resuscitation) → reoperative exploration.
- **Arteriovenous fistula.** Management depends on size, location, symptoms and overall status: observation; endovascular coil embolisation or stent; surgery for complex or symptomatic fistulas.

Concept — complications follow anatomy: the right side threatens duodenum and liver, the left side pancreas, spleen and the SMA, the upper pole the pleura; and after partial nephrectomy, the two late threats — urine leak and arterial bleeding — are each managed by escalating drainage or selective embolisation before any reoperation.

KEY POINTS

- **The right renal artery arises posterior to the LRV and the IVC.** In a difficult right radical nephrectomy it may be more accessible interaortocavally, posterior and medial to the IVC, near its aortic origin; cephalad retraction of the LRV as it crosses the aorta aids exposure.

- **The right gonadal and adrenal veins drain into the IVC; the left gonadal and left adrenal veins drain into the LRV.**
- **Simple nephrectomy — removal within Gerota's fascia — is for benign disease.**
- **Key elements of radical nephrectomy: early ligation of renal artery and vein, and removal outside Gerota's fascia; spare a radiographically normal ipsilateral adrenal when feasible.**
- **Left radical nephrectomy, especially for upper-pole masses: avoid ligating the SMA, which lies on the anterior aorta 1–2 cm cephalad to the LRV.**
- **Regional lymphadenectomy from renal hilum to aortic bifurcation is used in selected advanced local disease when feasible; its effect on progression-free and overall survival is controversial.**
- **Partial nephrectomy is the treatment of choice for stage T1 tumours when technically feasible, even without renal insufficiency.**
- **Goal of partial nephrectomy: complete excision with negative margins and maximal preservation of benign parenchyma — by enucleation or resection.**
- **Known IVC thrombus → intraoperative TEE after induction and intubation to define the distal extent before starting.**
- **Right radical nephrectomy with thrombectomy: the suprarenal IVC can be resected only if the LRV has been ligated on the caval side of its tributaries (gonadal, lumbar, adrenal), which then drain the left kidney.**
- **Left-sided tumour: the right kidney has no such tributaries, so the suprarenal IVC should not be resected unless alternative right renal drainage is provided — autotransplantation, or saphenous vein graft to the splenic, portal or inferior mesenteric vein.**
- **Alternatives to CPB include VVB and extensive liver mobilisation.**
- **IVC reconstruction: the lumen can be narrowed by half and closed primarily; larger resections need a pericardial patch or PTFE graft.**